PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME										
CT GROUP	1	6	S			INDEX NUMBER				

BIOLOGY **9744/01**

Paper 1 Multiple Choice **21 September 2017**

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

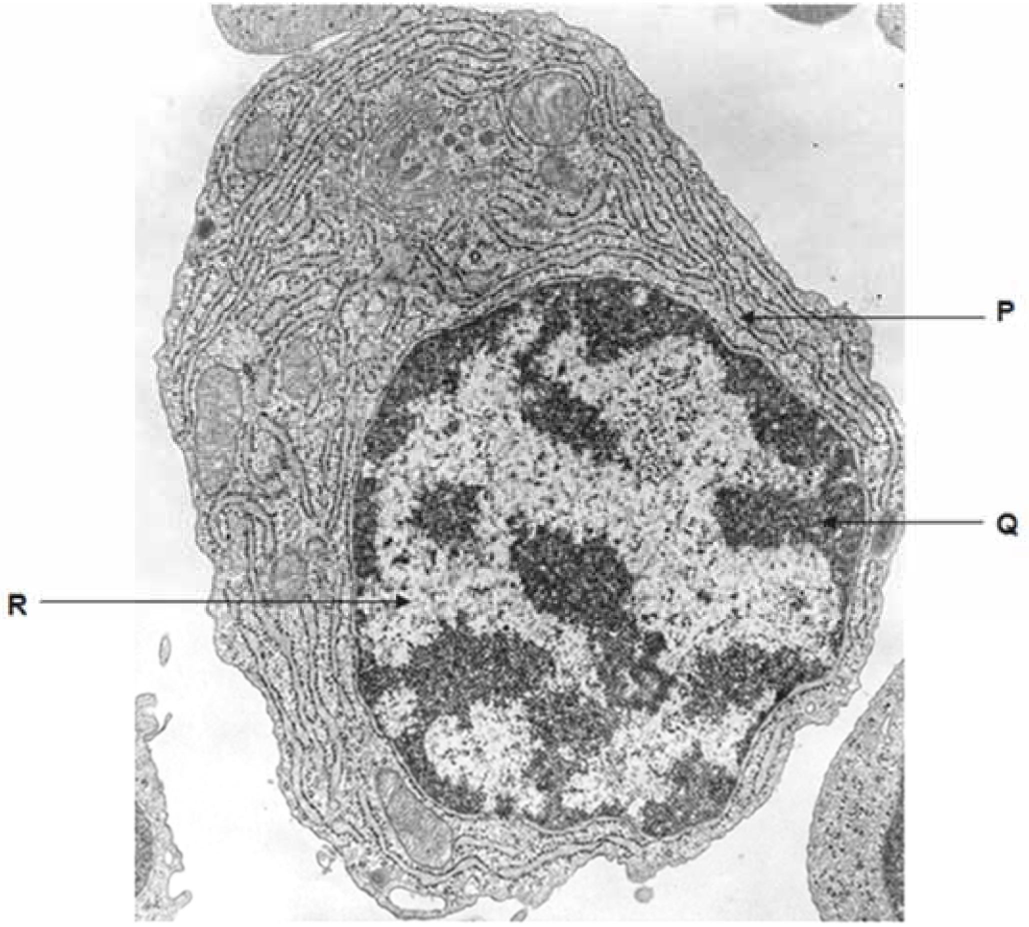
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, CT group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

1 The figure below shows an electron micrograph of a cell.



Which of the following about structures **P**, **Q** and **R** is correct?

	P	Q	R
A	provides large surface area for attachment of ribosomes	contains demethylated DNA	contains acetylated histones
B	transport of proteins to Golgi apparatus	histones are deacetylated	active condensation of chromatin
C	synthesis of phospholipids and steroid hormones	transcription of genes silenced	synthesis of proteins on free ribosomes
D	synthesis and processing of membrane proteins	contains methylated DNA	active transcription of genes

2 The following statements have been made about the cell theory.

1 All cells arise from pre-existing cells by division, with the exception of the first cell that came into existence.

2 All known living things are made up of more than one cell.

3 All cells contain hereditary information that is passed from cell to cell.

Which of the following statements about the cell theory are correct?

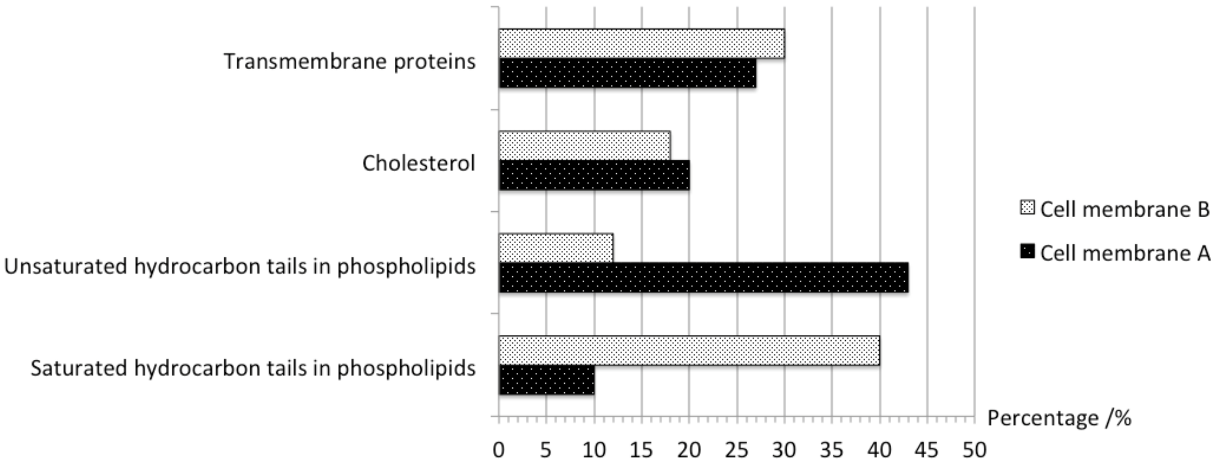
A 1 and 2

B 1 and 3

C 2 and 3

D All of the above

3 Reindeer are well adapted to survive extreme cold winters. One of these adaptations is the cell membrane composition at different parts of its body. The graph below shows the percentage composition of cell membrane components of cells A and B taken from two different parts of the reindeer's body.



Which of the following statement best explains the differences in the membrane composition in Cell A and Cell B?

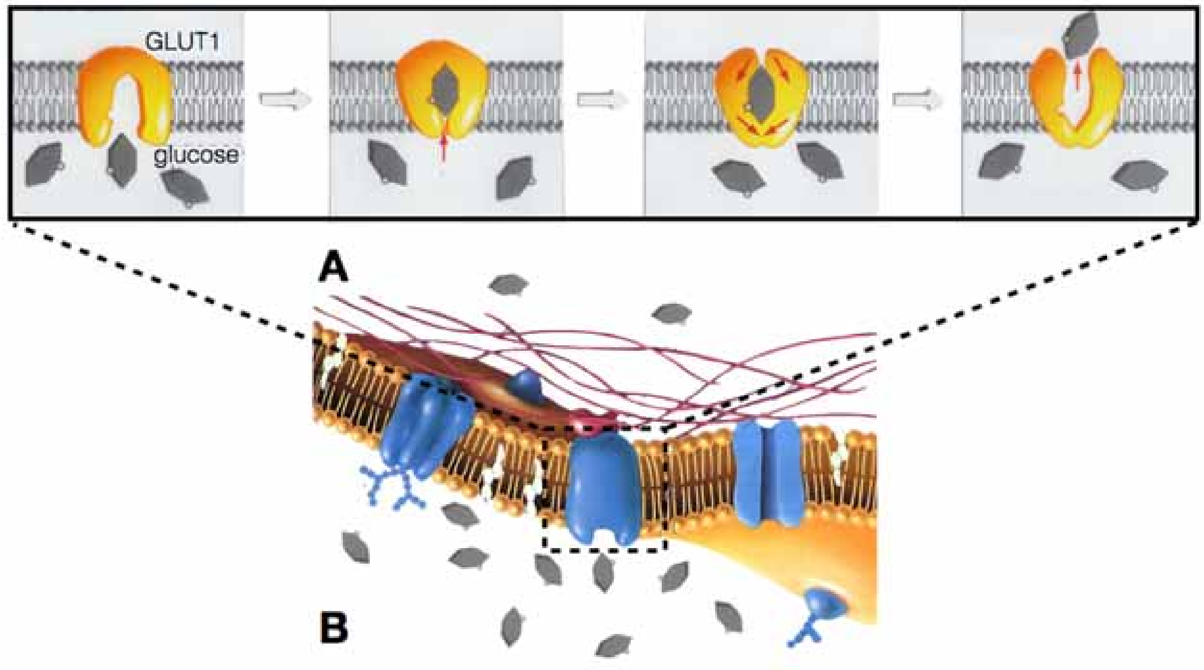
A Cholesterol decreases the membrane fluidity and prevents the membrane from breaking up by restraining the movement of phospholipids.

B Cell membrane A is taken from a lower part of the reindeer's leg as the unsaturated hydrocarbon tails will prevent the fatty acids from packing close to each other.

C Cell membrane B is taken from a lower part of the reindeer's leg as the saturated hydrocarbon tails will prevent the fatty acids from packing close to each other.

D Transmembrane proteins maintain the osmotic balance between the interior and exterior of the cell, hence preventing the cell membrane from solidifying at low temperatures.

4 Glucose Transporter (GLUT-1) is found in cell surface membrane of cells (denoted in dotted box) throughout the body and its mechanism in transporting glucose is illustrated in the following figure.



The following statements were made.

1 Glucose cannot diffuse directly across the membrane as it is a polar and large molecule.

2 There is net movement of glucose from intracellular to extracellular matrix through hydrophilic channel of GLUT-1.

3 GLUT-1 pumps glucose across the membrane via active transport.

4 GLUT-1 undergoes conformational changes that alternates between high and low affinity for glucose molecules.

Which of the following statements are incorrect?

A 1 and 4

B 1 and 3

C 2 and 4

D 2 and 3

5 Which of the following correctly describes starch, glycogen and cellulose?

	Starch	**Glycogen**	**Cellulose**
A	Can be easily hydrolysed to release α-glucose monomers	Angle of α 1,4 bonds and CH_2OH side chains results in helical chains	Made up of long linear chains due to 180^o rotation of alternate β-glucose
B	Insoluble due to the –OH groups projecting into the interior of helices	Made up of α-glucose monomers with 1,4 glycosidic bonds resulting in highly branched chains	Cellulose chains are organised into microfibrils and macrofibrils
C	Highly branched amylose due to α-1,6 glycosidic bonds	Highly compacted molecule that serves as a good energy storage in animal cells	Provides structural support and prevents cells from bursting when turgid
D	Highly compacted molecule that serves as a good energy storage in plant cells	Made up of β-glucose units with 1,4 glycosidic bonds resulting in highly branched chains	High tensile strength is due to the accumulative strength of the covalent cross linkages between cellulose chains

6 Compared to globular proteins, fibrous proteins are

A more resistant to high temperatures.

B less regular in structure.

C more readily soluble.

D more reactive chemically.

7 Which of the following statement about enzymes is correct?

A The high specificity of an enzyme is solely due to its specific 3D conformation.

B Before binding, the substrate must be exactly complementary to the active site of an enzyme in order for enzyme-substrate complexes to be formed.

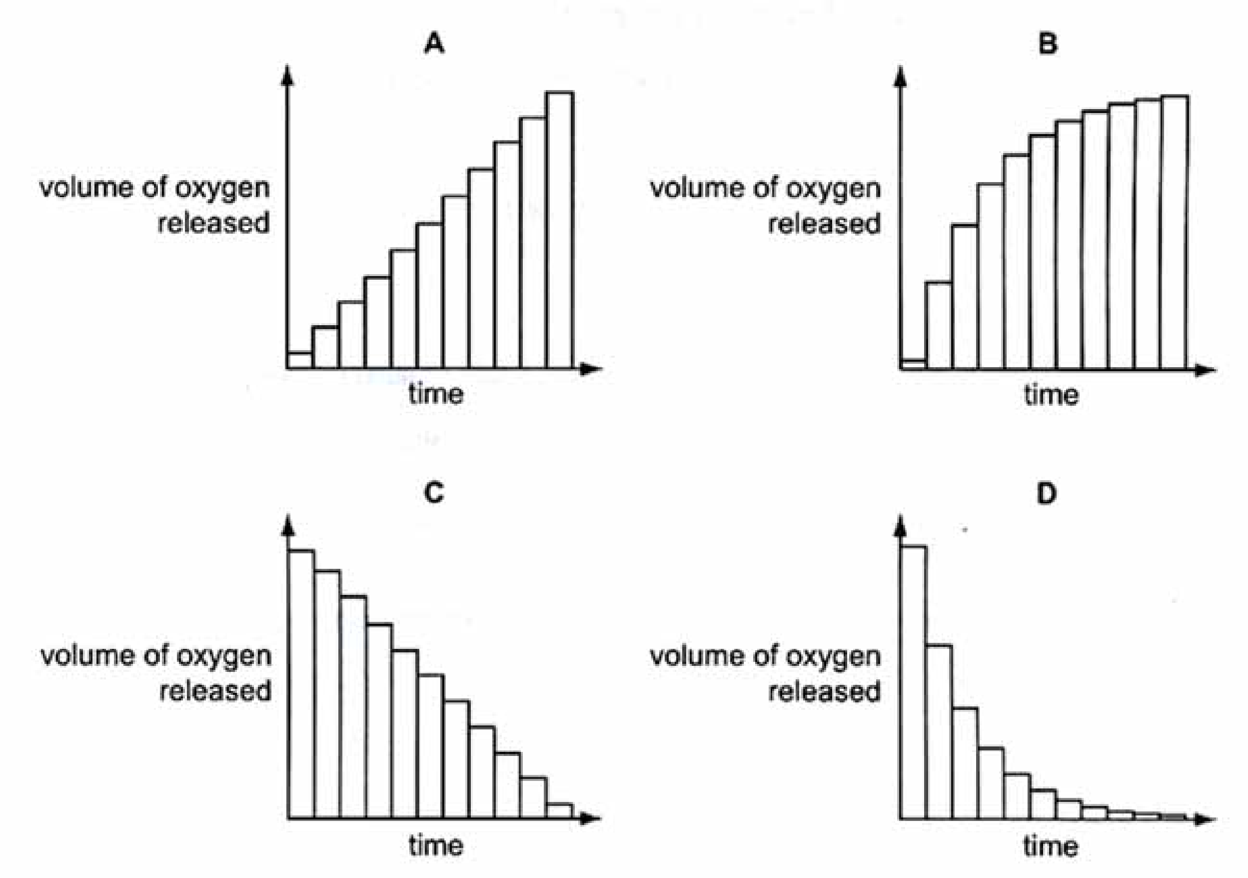
C Formation of enzyme-substrate complex serves to lower the activation energy of the reaction.

D The Michaelis constant, K_M, is the substrate concentration when rate of enzymatic reaction reaches maximum, V_{max}.

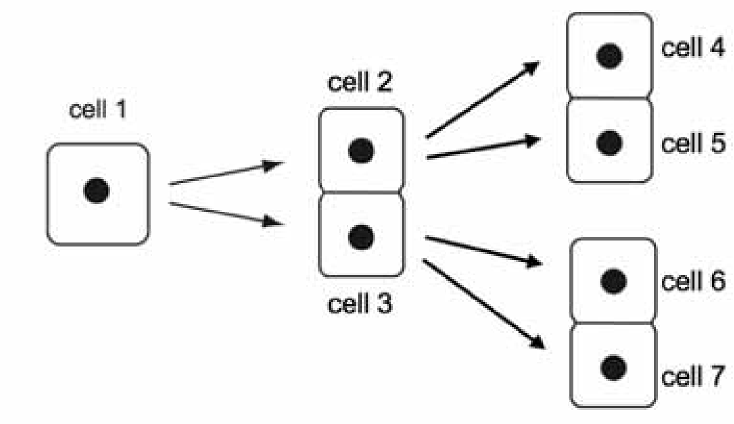
8 The decomposition of hydrogen peroxide to water and oxygen is catalysed by the enzyme catalase.

During an investigation, 2 cm^3 of catalase was added to 20 cm^3 of hydrogen peroxide and the volume of oxygen released was collected at intervals over a period of time.

Which bar chart shows the result of this investigation?



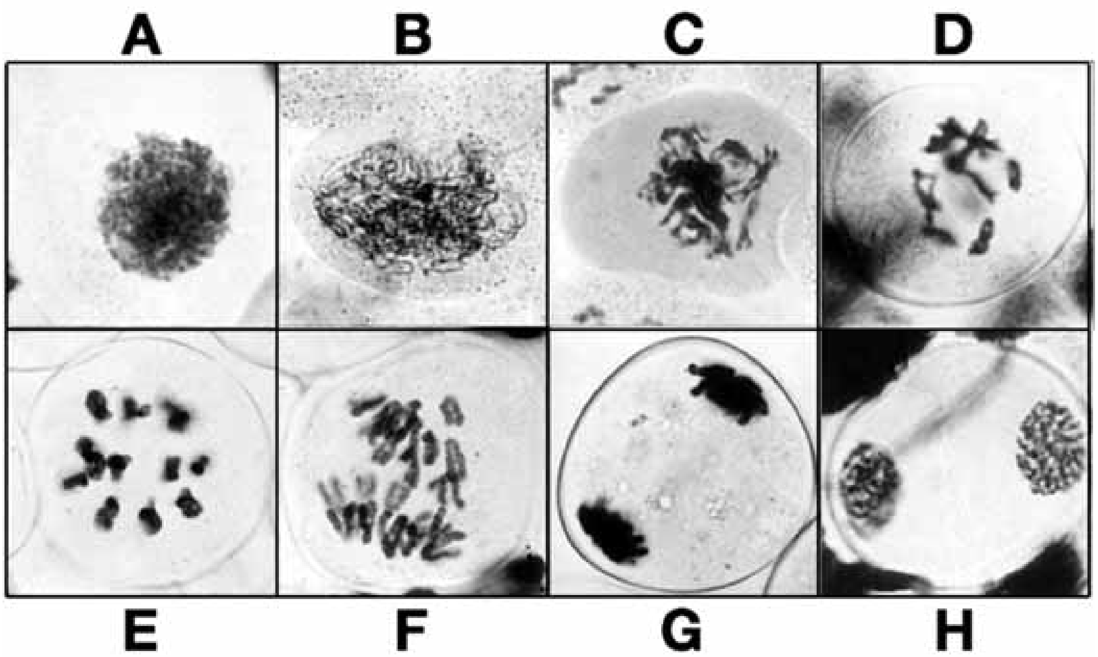
9 The diagram shows a plant cell (2n=18) at the end of prophase I of meiosis (cell 1), two daughter cells just after telophase I (cells 2 and 3) and four daughter cells just after telophase II (cells 4, 5, 6 and 7).



How many DNA molecules are there in the nucleus of cell 1, cell 2 and cell 4?

	Cell 1	**Cell 2**	**Cell 4**
A	18	18	9
B	18	9	9
C	36	18	9
D	36	18	18

10 The following diagrams **A - H** shows some stages in sequence during cell division in *Lilium grandiflorum* (Lily).



Which of the following statements best describes the indicated stage(s)?

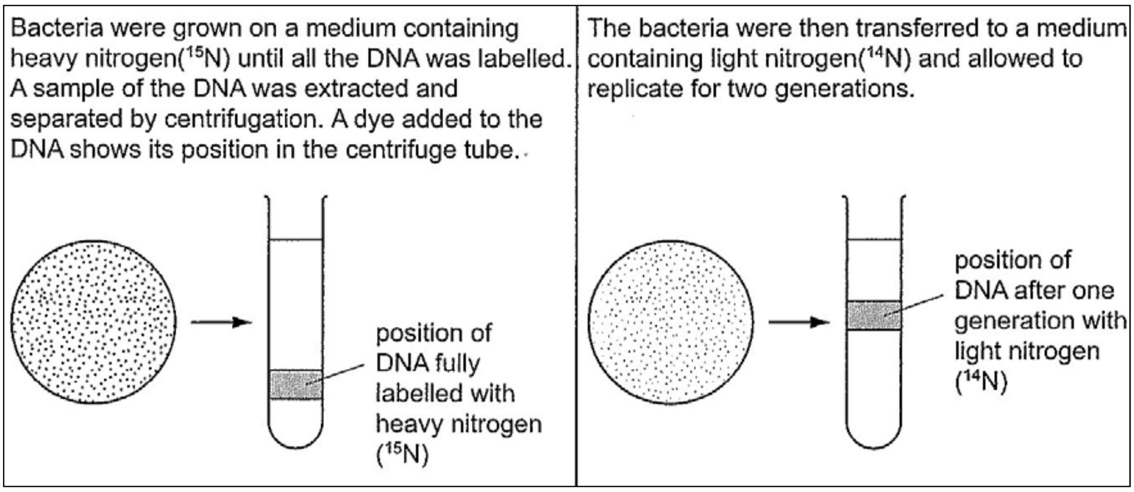
A In stage **A**, condensation of chromatin occurs as centrioles migrate to the opposite poles.

B In stages **C** & **D**, chiasmata are formed and crossing over takes place.

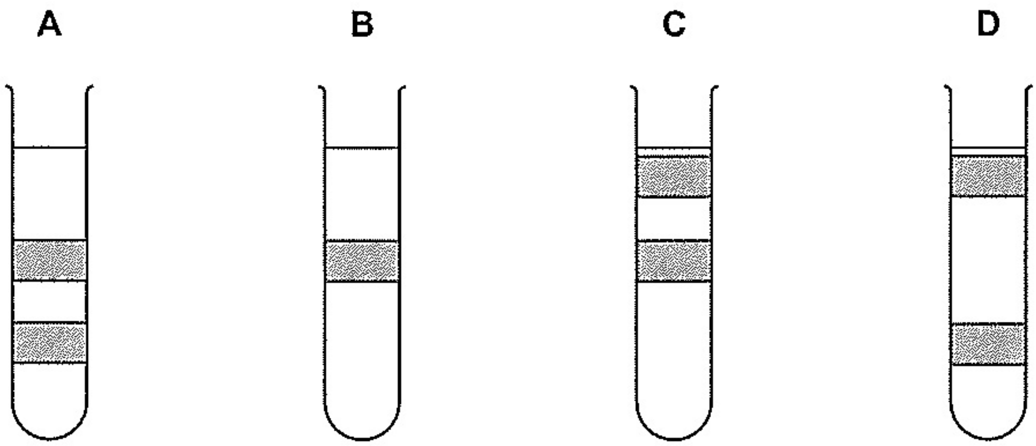
C Stage **E** shows the alignment of 11 chromosomes along a metaphase plate.

D In stage **F**, sister chromatids separate and migrate towards opposite poles.

11 The diagrams show an investigation into semi-conservative replication of DNA.



Which tube shows the position of the DNA after two generations of semi-conservative replication in liquid nitrogen (^{14}N)?



12 The table below shows a list of characteristics displayed by mutant strains of *E. coli* during DNA replication and the possible reasons.

No	Characteristics	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed.	DNA ligase activity is missing.
2	Supercoils are found to remain at the flanks of the replication bubbles .	DNA helicase has a low activity.
3	Synthesis is very slow.	DNA polymerase keeps dissociating from the DNA and has to re-associate.
4	No initiation of replication occurs.	A-T rich region at origin of replication deleted.

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

A 2 and 3

B 1 and 4

C 1, 3, and 4

D All of the above

13 Which of the following statements correctly describes the genetic code?

1 It is degenerate as there are three codons that act as stop codons, which stops the generation of polypeptide during translation.

2 The codons in the genetic code do not overlap, and are read as distinct reading frames during translation.

3 The genetic code is a triplet code, except in prokaryotes where the bases are read in doublets due to the smaller 70S prokaryotic ribosomes.

4 It is possible that 1 amino acid can be coded for by more than one triplet code.

A 2 and 4

B 1 and 3

C 2 and 3

D 1 and 4

14 If the *lac* operon were to unable to produce any enzymes regardless of the presence or absence of lactose, what could be the likely reason for this?

A *LacI* has been deleted.

B Promoter sequence has been deleted.

C Repressor is unable to bind to the operator.

D Lactose is always bound to the repressor.

15 Which of the following statement(s) regarding viral reproductive cycle is true?

1 All enveloped viruses contain enzymes embedded in their membranes that facilitates the release of viral progeny.

2 Orthomyxoviruses carry RNA-dependent RNA polymerases that allow the synthesis of a complementary DNA from its positive strand single-stranded RNA

3 HIV is an RNA virus that carries a 2 positive strand single-stranded segmented RNA, which acts as a substrate for reverse transcriptase

4 all viruses complete their maturation by budding from the host cell

A 1, 2 and 4

B 2 and 4

C All of the above

D None of the above

16 Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were unable to produce the protein, Rb.

Addition of Rb to these cells reduced their rate of division.

What can be concluded from this investigation?

A Both chromosomes in the cancer cell carry alleles for tumour suppressor gene.

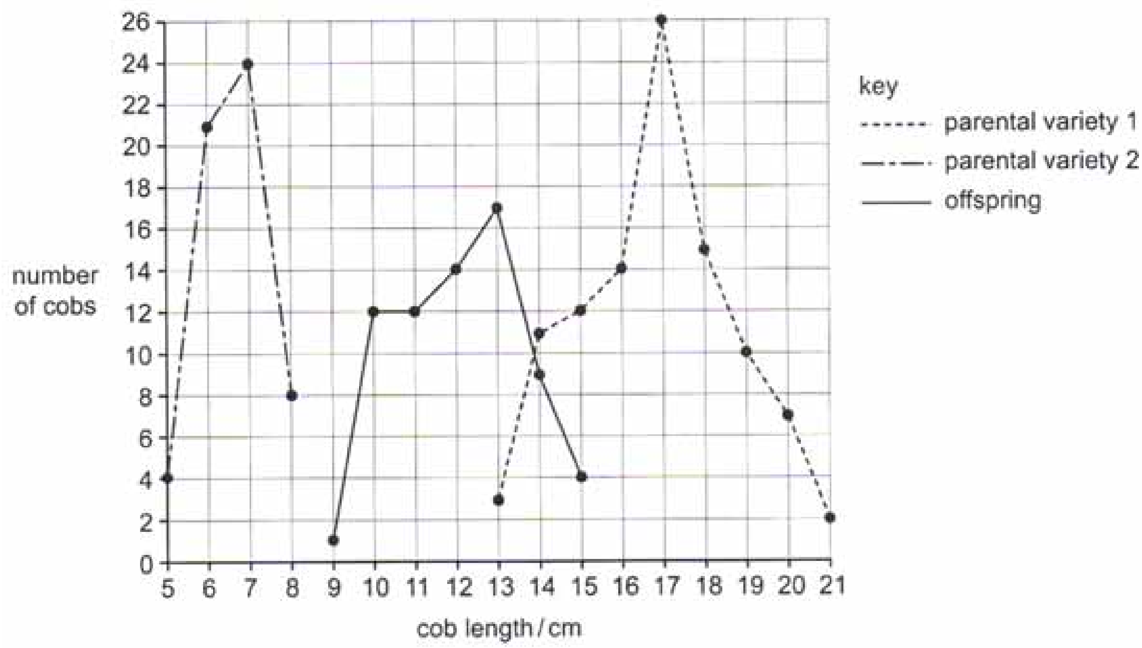
B Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.

C Both chromosomes in the cancer cell carry alleles for proto-oncogene.

D Both chromosomes in the cancer cell have the allele for proto-oncogene deleted.

17 Two pure-bred lines of two varieties of maize which differed markedly in cob length were crossed. The length of the cobs by the two parental varieties and their offspring were measured to the nearest centimetre. The number of cobs in each length category was counted.

The graph shows the results.



Which is the cause of the phenotypic variation shown in cob length within the two parental varieties and their offspring?

A segregation and independent assortment of alleles

B linkage and crossing-over at meiosis

C additive effect of different genes

D various environmental factors

18 The coat colour of Labrador retrievers is controlled by two genes, **B/b** and **A/a**. Allele **B** codes for black coat, while allele **b** codes for brown coat. The coat colour of a Labrador retriever with a genotype **aa** is yellow.

A cross between a male black Labrador retriever and a female yellow Labrador retriever produced some black puppies and yellow ones.

What are the genotypes of the parental dogs?

	Black retriever	**Yellow retriever**
A	AaBb	aabb
B	AaBb	aaBb
C	AaBb	aaBB
D	AABb	aaBb

19 The table below shows the results of a series of crosses in a species of a small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

A one gene with a pair of co-dominant alleles

B one gene with multiple alleles

C sex linkage of the allele for grey coat colour

D two genes, each with a dominant and recessive allele

20 Which of these statement(s) is/are true?

1 Royal jelly contains an active ingredient that when fed in alternation with another diet to bee larvae will result in the development of queen bees.

2 The Himalayan rabbit has the genotype for black fur all over its body, but the enzyme that produces the black pigment is temperature sensitive.

3 Height is a polygenic trait, which can be reduced by poor nutrition.

4 The effect of individual polygenes cannot be observed, but the additive effect can be observed.

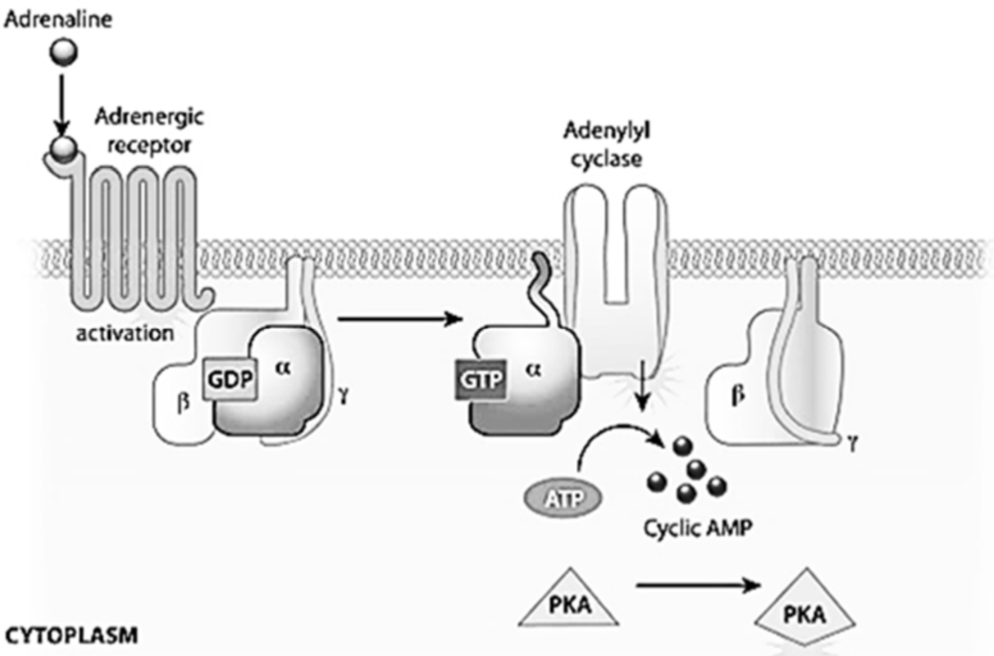
A 2, 3, and 4

B 1, 2, and 4

C 1 and 3 only

D 2 and 4 only

21 The following diagram shows the activation of the G protein-coupled receptor (GPCR) by the binding of adrenaline to the receptor. A mutation leads to constitutive signal transduction.



Which of the following is a possible explanation of the mutation?

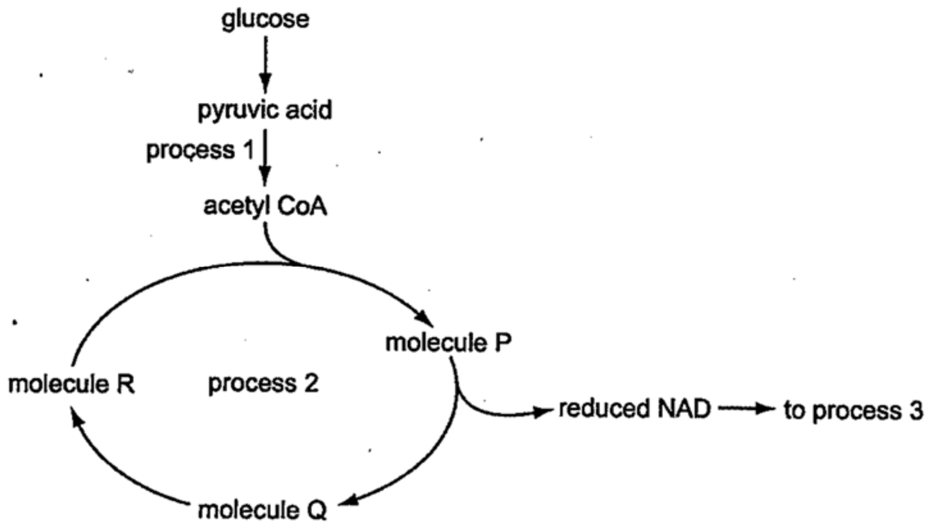
A Conformational change in adenylyl cyclase such that it cannot convert ATP to cyclic AMP.

B Adrenaline cannot bind to the receptor.

C Cyclic AMP cannot bind to PKA.

D GTPase in G protein fails to hydrolyse GTP to GDP.

22 The diagram shows a summary of aerobic respiration.



Which statements are correct?

1. Process 1 occurs in cytosol and process 2 occurs in mitochondrial matrix.
2. Process 2 and 3 occur in mitochondrial matrix.
3. Process 1 occurs in mitochondrial matrix and process 3 occurs in inner mitochondrial membrane.
4. Process 1 produces 2 ATP and 2 NADH per glucose molecule oxidized.
5. Process 2 produces 2 ATP, 10 NADH and 2 FADH2 per glucose molecule oxidized.
6. Process 3 is responsible for producing about 90% of the total yield of ATP from the hydrogen carriers reduced per glucose molecule oxidized.

A 2 and 4

B 1 and 5

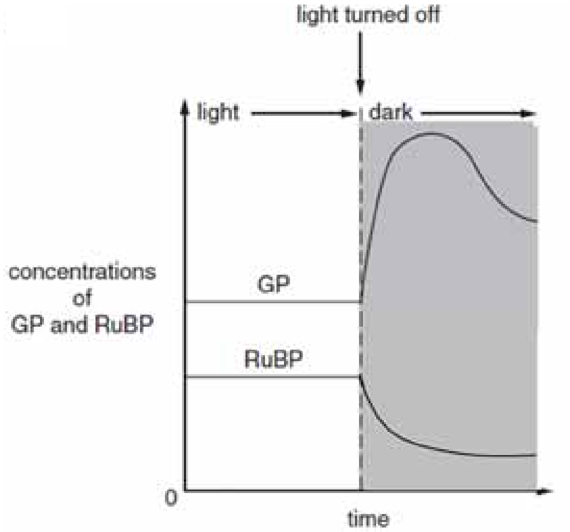
C 3 and 6

D 1 and 6

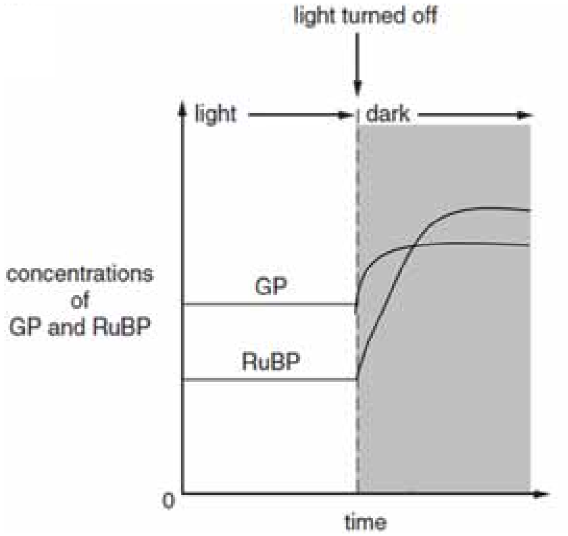
23 Concentrations of glycerate-3-phosphate (GP) and ribulose bisphosphate (RuBP) were measured from samples of actively photosynthesising green algae in an experimental chamber.

Which of the following graphs show how the concentration of these compounds changed when the light source was turned off?

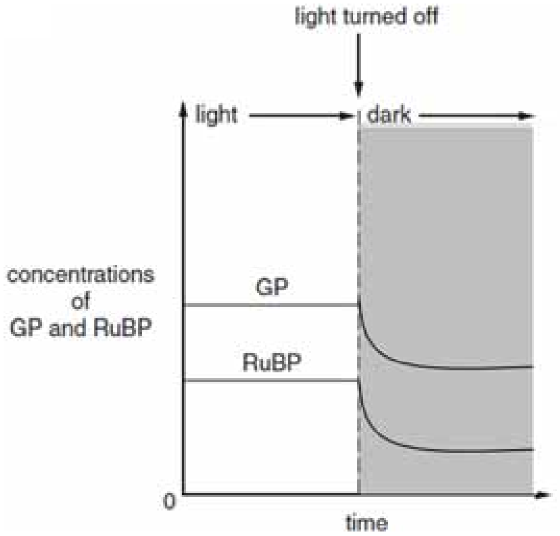
A



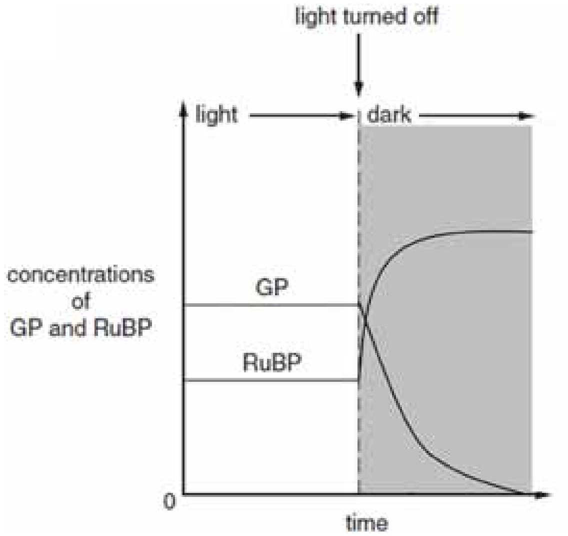
B



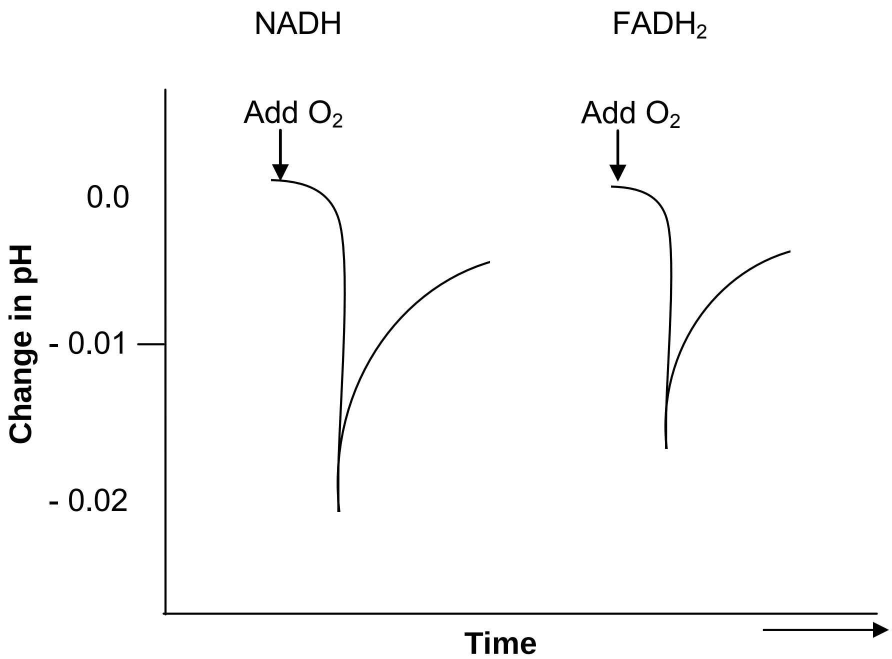
C



D



24 Isolated mitochondria were incubated with NADH in one experiment and an equal amount of FADH2 in another experiment. The mitochondria were initially deprived of oxygen. The pH of the intermembrane space was then monitored as a known quantity of oxygen was added. The results are shown in the graph.



Which of the following can be concluded based on the results?

1 Upon the addition of oxygen, glycolysis and subsequently, link reaction, Krebs cycle and oxidative phosphorylation occurred.

2 Electron transfer was initiated by the addition of oxygen.

3 The pH drop was greater with NADH than with FADH2, which is consistent with the greater ATP yield that accompanies the oxidation of NADH.

4 The rapid decline in pH indicates that protons were pumped into the intermembrane space when oxygen was available.

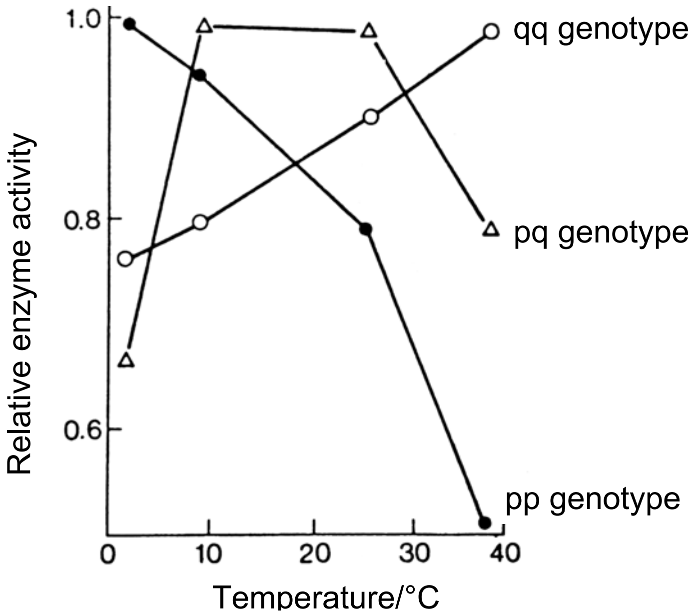
A 1 only

B 2 and 4 only

C 2, 3 and 4 only

D All of the above

25 In the North American catfish *Catostomus clarki*, two alleles, represented by p and q, control the synthesis of a vital enzyme. The three possible genotypes (pp, pq, qq) lead to the synthesis of variations of the same enzyme with different optimal temperatures as shown in the graph below.



When the mean annual temperature is 5°C, which of the following statements is correct?

A Frequency of allele p in the gene pool will increase.

B Frequency of allele q in the gene pool will increase.

C Allele p will become dominant and the allele q will become recessive.

D The heterozygotes will have an advantage over the homozygotes.

26 Which of the following statements correctly relate to molecular phylogenetics?

1 Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.

2 Organisms with similar base sequences in their DNA are closely related to each other.

3 The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.

4 The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

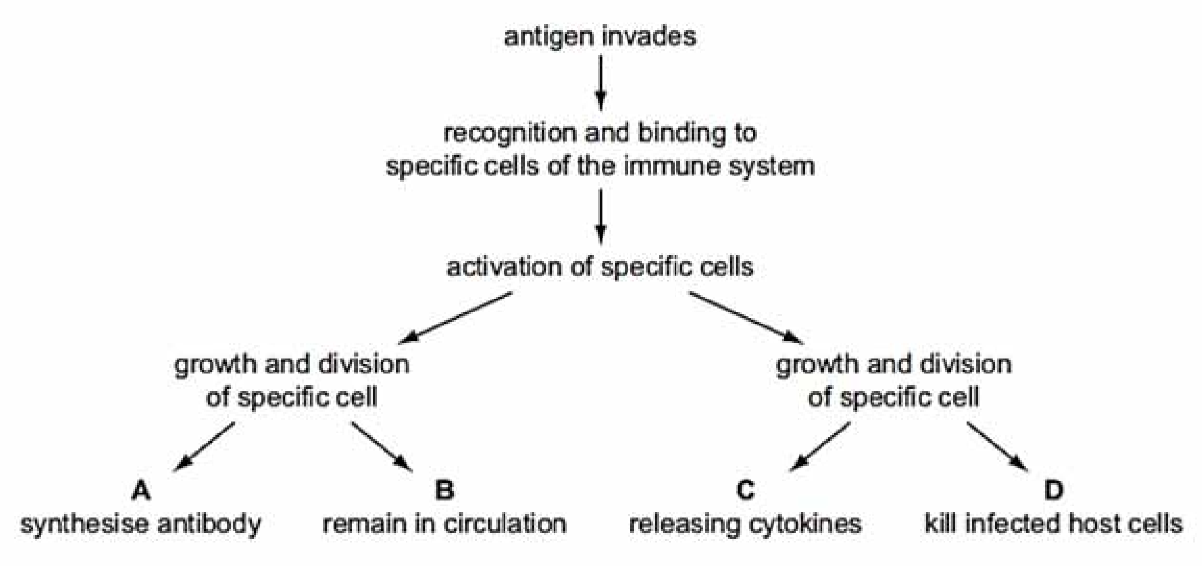
A 1 and 2

B 1 and 4

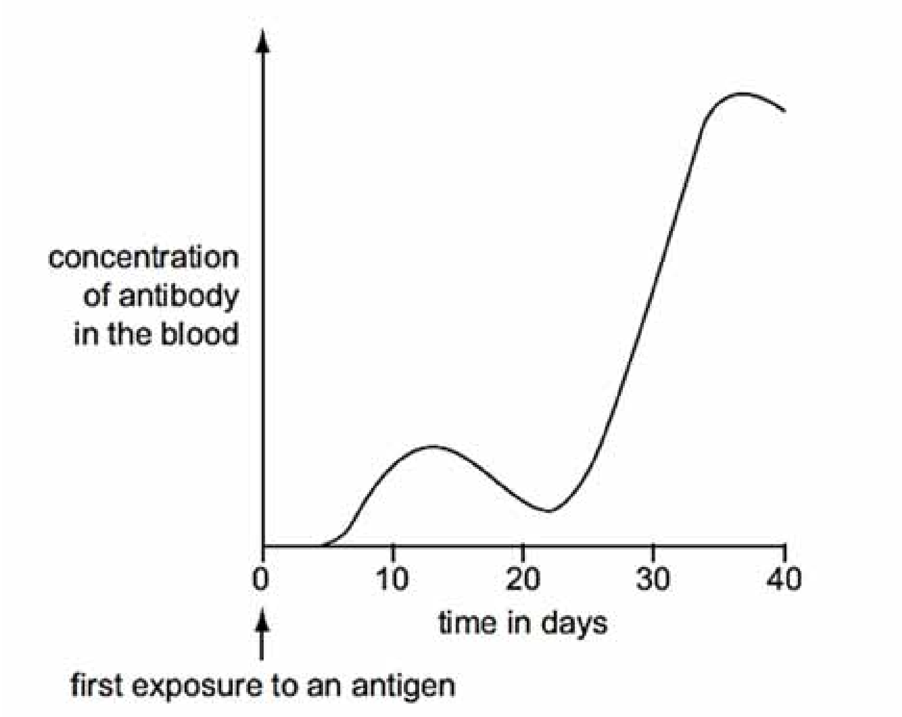
C 2 and 3

D 3 and 4

27 Which describes a T-helper lymphocyte?



28 The graph shows the amount of antibody produced in response to an antigen.



From the graph, which statement is correct?

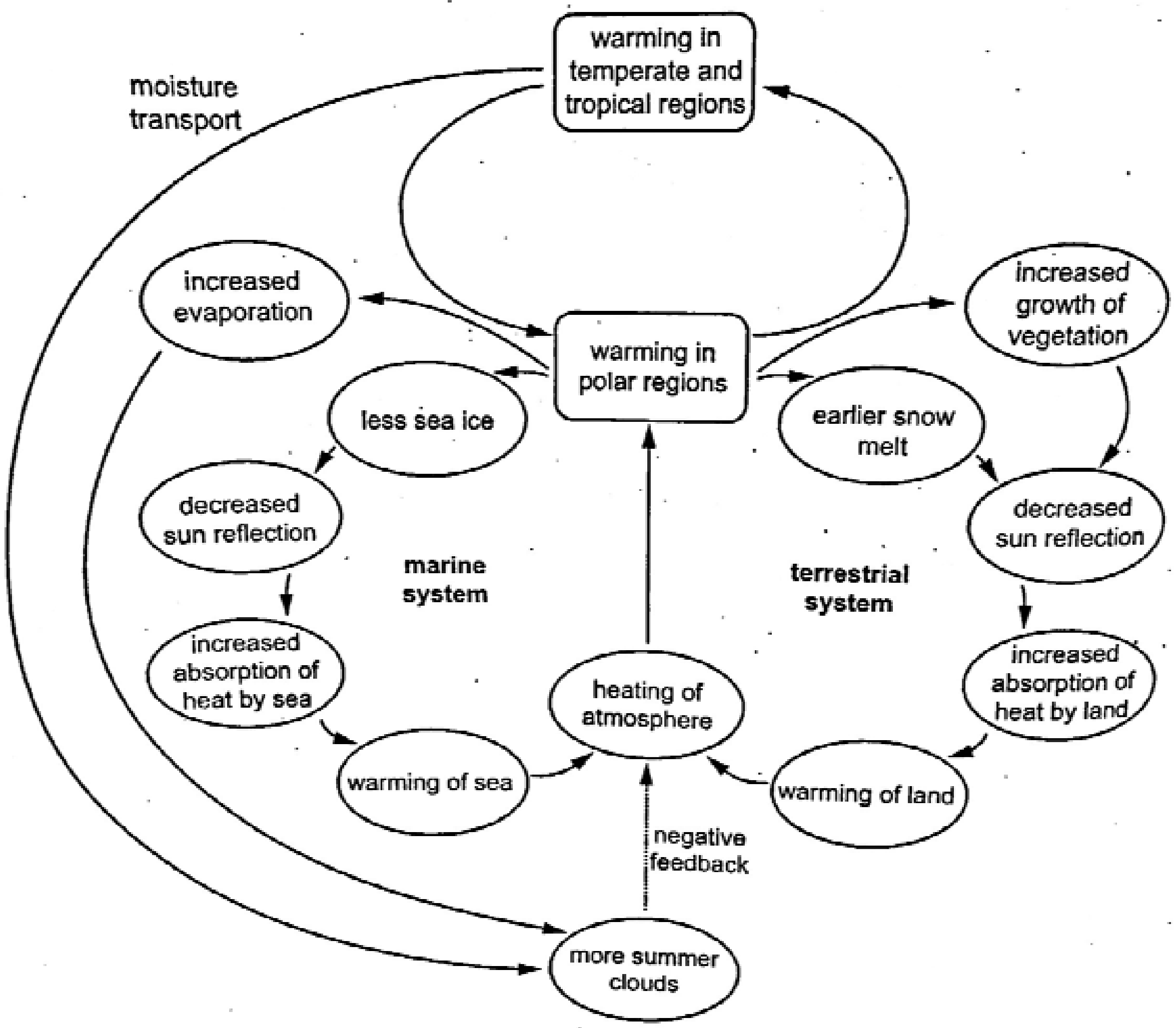
A It takes 25 days to achieve active immunity.

B Memory cells for this antigen are present in the body within 20 days.

C T-helper lymphocytes are activated on day 12.

D The second exposure to the antigen occurred on day 25.

29 The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

A Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.

B Increased evaporation leads to more rainfall, which absorbs heat from the land and the sea.

C Melting sea ice causes more cloud formation, which increases absorption of heat in the atmosphere.

D Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.

30 Why is it difficult to control the spread of malaria?

1. There is an increase in host range beyond mosquitoes that can contribute to the spread of malaria.
2. The mosquito vector rapidly evolves resistance to insecticides.
3. The plasmodium pathogen shows great antigenic variability.
4. Civil unrest and poverty results in overcrowded living conditions.

A 1, 2 and 3

B 1, 2 and 4

C 2 and 3

D 3 only

END OF PAPER

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